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Cybernetics, or the metatechnology of a machine

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ABSTRACT
In 'Cybernetics, or the metatechnology of a machine', Max Bense responds to the dramatic developments in communications technology, information theory, and computation in the late 1940s. Published in March 1951, the short essay focuses on the 'Eniac' as a cybernetic machine and argues that the invention of such machines proves that technology implements logical principles – a first step in his ontological theory of machines. In eight short sections, Bense offers brief historical insights into the philosophical, mathematical, physical, and cybernetic foundations of the relationship between humans and machines, concluding with a prognosis for a future in which human existence can only fully be understood as technical existence.

KEYWORDS
Cybernetics; ontology; computing

Philosophy

We understand philosophy here as a group of sciences that belong neither to the natural sciences nor to the humanities, but which are indispensable to the development of both. They investigate the foundations, domain, and internal coherence of positive sciences like mathematics, physics, biology, literary history, law, and so forth. In particular, we count logic, epistemology, theory of science, ontology, and metaphysics among these philosophical sciences. In and of themselves they have no independent object; they get their object, instead, from the positive sciences, transform it, transpose it to their level, and occupy themselves with characteristics whose relevance or irrelevance cannot be determined by the methods of the non-philosophical sciences. Philosophical sciences, to be precise, speak exclusively about non-philosophical, positive, *actual sciences*; without these latter, the former are – so to speak – 'empty'. Philosophical sciences are therefore meta-sciences. And in fact we also count metaphysics and metamathematics among them; in a certain sense both Aristotelian metaphysics and also Hilbertian metamathematics correspond to the concept of the philosophical sciences we have laid out above. But recently it has become ever clearer that this group of philosophical sciences can be derived not only from literature and art, but from technology as well. Its object, too, can be raised to the plane of philosophical contemplation. And it seems that this must be done in order for us to cope intellectually with technology. The philosophical construction of a discipline of metatechnology is therefore reasonable.

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Philosophy of nature and technology

Nature has always been an eminent occasion for the philosophical cultivation of thoughts. The scientific philosophy considered here receives its object, which we call 'nature', from the modern natural sciences. This 'nature' of physics, chemistry, biology, and so forth, is, after Galileo, no longer simply 'nature as given'. It is 'changed nature', the nature of the laboratory and of the thought experiment. The free fall of Aristotle was still the fall of the leaves and the fall of snow in a free nature; Galileo's free fall was an analytically traced fall in airless space, one which could be produced in a vacuum chamber. The free fall of the Greek was natural; the free fall of the great Tuscan was artificial, denatured. Two things were necessary so that Galileo could formulate his famous laws: first, an abstract contemplation of a mathematical kind, which would allow him to trace the numerical interdependence of two variable quantities, falling time and falling distance, for example. And this abstract thought experiment occurs in connection with a geometrical figure, as the *Discorsi* of 1638 demonstrate.[1] This experiment contains certain beginnings of infinitesimal and coordinate-geometrical thinking, the details of which are not relevant here. Thus nature was first reflected in geometrical form. Then comes the second aspect, the experiment, the test in reality. And now it becomes clear that Galileo is not testing the mathematical law of nature on nature as given at all, but instead on nature in the laboratory, on the machine. He constructs an inclined plane with carefully chosen stopping points and with a groove which a ball can run down. A precise water-clock allows him to measure the time and distance of the ball running down the inclined plane. Mathematical physics and technological physics intersect here. They provide philosophy with a second, artificial, at once abstract and technological nature for contemplation. Today, we know that we cannot proceed in the philosophy of nature from nature as given; we need the nature of physics and technology. The philosophy of technology has become a constituent part of the philosophy of nature.

The Galilean deduction

But the Galilean deduction not only shows whether and how natural processes can be described in their quantitative development through the use of mathematical formulas. It also shows whether and how thought processes – the consideration of the triangle – can be translated into technological processes – the path that the ball takes down the inclined plane. Thus the famous passage from the *Discorsi* that we always have in view here is, philosophically speaking, also an example of the realisation that one and the same process, when considered broadly enough, can be reproduced equally well in the form of *thought* or of *technology*. Here the mysterious double nature of the technological world, as well as of technological intelligence – its being in equal measure pure theory and pure practice, pure thought and pure material, abstract and concrete – enters into methodological consciousness for the first time. Technology belongs to the domain of *ideal*, but also to the domain of *real relations of being*, and thus to both the subjective and the objective spheres of being. Ontologically speaking, this means that *technology reaches further than nature*: it encompasses the areas of both ideal and real being. But

[1]Galileo Galilei, *Discorsi e dimostrazioni matematiche intorno a due nuove scienze*, 1638 (German 1891).

this means that it is possible at least in principle, as a consequence of the theory of being as such, so to speak, to think about the technological reproduction of ideal, mental processes, i.e., about the material, mechanical construction of thought processes such as addition, multiplication, but also logical operations. Pascal's calculator[2] was just as inevitable in the domain of technology as Leibniz's calculus was in the field of pure theory.

The initial question is always the same: Which technological process corresponds to a particular mental process? This reduction to identical formal operations makes it easier to find the technological analogy for an act of the intellect. When we have found the formal process, we gain access to both its mechanical and its mental reproduction. For instance, we can think of a *proposition* – i.e., the entity discussed by logicians that has the property of being true or false – as a relay, a switch that can be turned off or on. The logical binarism of the proposition, which can be true or false, a purely formal attribute, corresponds to the technological binarism of being on or off. The logician is additionally concerned with such things as the conjunction 'not only … but also', which connects two sentences. With the help of this conjunction, two simple propositions can be combined. Assuming that the proposition is a linguistic entity that can be true or false, as stated, then the combination of two propositions created by the conjunction 'not only … but also' is also true or false, depending on whether the conjoined propositions are true or false, or one is true and the other false. Clearly there are four cases to be distinguished. The logician can capture these four cases very precisely, as it were, in a specially constructed table known as a matrix. There are specific matrices for individual conjunctions such as 'not only … but also', 'and', 'or', 'if … then', and so on. The matrix expresses the mechanism of conjunction in each case. If we then ask whether there is a technological process, a machine, that functions in accordance with the logic of the conjunction 'not only … but also' as applied to two propositions that can be true or false, then the answer will point to the Hollerith machine.[3] With the help of the Hollerith machine it is possible, for example, to select from a card file all of the physicists who are also bacteriologists, should that question be posed. The 'not only … but also' logic is easy to follow. Roughly put, then, a Hollerith machine repeats the mechanism of the 'not only … but also' matrix found in propositional calculus. Like mathematics, then, logic also carries out certain processes that can be technologically reproduced in a straightforward manner. Computing machines and Hollerith machines demonstrate this. But as far as the process of counting is concerned, it is not necessary to count the numbers with the help of a decimal system.

The series of natural numbers, 1, 2, 3 … can also be expressed with the help of two signs, that is, two numbers, in a dyadic system. And the two signs? They are produced through the use of the relay. The relay that is closed – turned on – corresponds to one number; the relay that is opened – turned off – to the other. This controlled relay expresses the series of numbers by way of the opening or closing of a contact: thus a

[2]Blaise Pascal developed his calculator and adding machine between 1640 and 1645. It was the oldest preserved machine until Bruno von Freytag-Löringhoff repaired the first German calculator for addition and subtraction by Wilhelm Schickhardt, which had been partly preserved. It was recently placed on display at the Kepler Museum in Weil der Stadt. Pascal's machine inspired Leibniz to design a calculator that could not only add and subtract but also multiply and divide. Built in 1693, it is now housed in the State Library of Lower Saxony in Hanover.

[3]In 1890, Hermann Hollerith built a sorting machine using punch cards and carriage control tape, based on the ideas of the British mathematician Charles Babbage from 1832. In 1896, he founded the Tabulating Machine Company in New York, which was later renamed International Business Machines (IBM).

contact can count. And with that, we have already entered into the realm of the great modern mathematical machines, and in a certain respect such machines consider mathematics in terms of processes, not of theorems. This, again, is a fact that corresponds to the findings of modern logic. Researchers such as H.B. Curry, among others, have recognised that the mathematical character of mathematics lies in its deductions, not in its putative objects.[4] In light of this insight, we can grasp that the technological process of *control*, that is, the precise, rhythmical regulation of mechanical processes in space and time, must be decisive for the mechanical reproduction of certain intellectual actions of our consciousness. And if these intellectual actions, additions, multiplications, counting, sorting, etc., become more refined, then the technological control processes must also become more subtle. The familiar mechanical control of reproducible processes is then replaced by electromagnetic and electronic controls, which have become familiar, e.g., in the simplest case of the standard control grid – the grid that controls the stream of electrons emitted by the cathode within the vacuum tube.

The cybernetic expansion

With that, we have arrived at the theory that Norbert Wiener – an American mathematician who also studied in Göttingen and is now a professor at the Massachusetts Institute of Technology – recently published under the title *Cybernetics – or Control and Communication in the Animal and the Machine* in 1949.[5] 'Cybernetics' comes from a Greek word whose stem can be translated as 'steering'. *Gubernator* is the Latin word for 'steersman'. The subtitle indicates that Wiener places modern electrical control technology and signalling – *Control and Communication* – but especially electronic computing machines, in the center of his reflections, which aim to determine certain identical processes in men and machines that can be technologically isolated. The incorporation of modern logic, mathematics, physics, psychology, physiology, psychiatry, anthropology, sociology, and metaphysics makes the work an essential book of the modern philosophy of nature and technology, a paradigm of what we would like to call *metatechnology*. Until now, technology was primarily a phenomenon pertaining to the surface of the inhabited and habitable sphere: what is taking shape now before our eyes is *depth technology*; we experience the way that it penetrates the fine structures of the world, the immaterial components, where its pathological features are consequently more concealed and more dangerous. The cybernetic expansion of modern technology thus means its *expansion under the skin* of the world; technology can no longer be considered as isolated (objectified) from the process of the world and its sociological, ideological, and vital phases. It incorporates everything, it has taken on an increasingly all-consuming character. Literature, art, music take on its features, just as at least since Galileo, science, medicine, and architecture, and at least since the Enlightenment, social and political processes have adapted to its structures.

[4]Haskell B. Curry, an American mathematician and logician who wrote his dissertation under the supervision of David Hilbert in Göttingen, published a book with the title *Outlines of a Formalist Philosophy of Mathematics* in 1951; excerpts of the project, which was conceived in 1939, had been published earlier.

[5]Norbert Wiener's book *Cybernetics* has been published in numerous editions and translations around the world since 1949. This work marked the beginning of the computer age.

This penetration of technology into zones that were previously out of reach, and its conversion into metatechnology, which constantly blurs the line between the so-called material and non-material realms, can be demonstrated by the fact that the cybernetic machines that Norbert Wiener discusses were not only created through the constant collaboration of logicians, mathematicians, and technicians (as the introduction to his book demonstrates in its history of programs and ideas) and based on the translation of logical and mathematical processes into technological procedures, as mentioned above, but they also exhibit functions corresponding to certain characteristics of human consciousness, such as the memory. I said 'functions of the memory', not 'memory'. That means that I merely want to say that cybernetic machines reproduce procedures whose formal execution corresponds to the procedure according to which we can understand the formation of memories in human consciousness. There is no need to be alarmed. Anyone familiar with psychoanalysis knows that there is a mechanics of our vital, emotional, and intellectual life that regulates repressions, complexes, sublimations of sexual symbolism in dreams, etc., and that offers great therapeutic value to those who understand it, as Freud and his school have shown. Once we have understood that the functions of the memory consist in the storage of signals, this technological manner of speaking immediately makes it easier to transition from a process based in consciousness to one based in technology. Henri Bergson, who distinguishes between two types of memory in his famous 1896 book *Matière et mémoire*, would now explain that only the repetition of an automated function by the consciousness – the first sort of memory – but not the formation of memory images – the second sort of memory – can be reproduced technologically.[6] In Franz Kafka's *The Trial*, there is a sentence: 'The key turning in the lock reminded K. that he had intended to stay only a short while'. The memory at work here is doubtlessly of the first sort. Marcel Proust's powerful title *A la recherche du temps perdu* seems to refer to memory of the second sort, which he identifies as 'mémoire involontaire', since it is 'a matter of chance'. By the way, I would like to note here that in the way that Balzac conceives his *comédie humaine* as at once a literary and a sociological program, and in the way that Proust in his cycle perceives the *temps perdu* as at once a literary and a metaphysical program, an indication of the penetration of metatechnological structures into the novel can be seen. The explication of Proustian memory images that Walter Benjamin offered in 1939 with the help of psychoanalysis supports this interpretation.[7]

But let us return to cybernetics. It is most likely, as Bergson demonstrates, that matter has no memory. But there are controllable, subtle electronic processes in vacuum tube assemblies that can reproduce all aspects of the memory that rely on the fine mechanics of signals and that fall within the domain of signal transmission, of *communication*, as Wiener says. The sensational chapter of his book, 'Computing Machines and the Nervous System', spares us no surprise in this regard. In electronic computers and

[6]Henri Bergson's *Matière et Mémoire, essai sur la relation du corps à l'esprit* was published in Paris in 1896 (German: *Materie und Gedächtnis*, 1907). As the leading representative of the French philosophy of life, he had a great impact in Germany, England, and America, as well. In 1927, he received the Nobel Prize for Literature. His influence on Marcel Proust is justifiably contested by Ilse Walther-Dulk in 'Auf der Suche nach der Philosophie Marcel Prousts' (*Semiosis*, Vol. 22/23, Issue 85–90, 216–231).

[7]Walter Benjamin's essay on Proust appeared in 1939 in Paris in the *Zeitschrift für Sozialforschung*, to which Benjamin contributed from 1932 to 1941.

analysers, the storage of intelligence signals corresponds to the work of memory in Bergson's terms, at least of the first sort, if only for the short periods of time while it is functioning. Imagine for a moment a frozen earth, devoid of humankind, and somewhere, safely preserved, in an ideal, durable museum, so to speak, all of the instruments that humans once possessed. Including a perfected *computer*, but no book, no written text. Then we could imagine that a portion of abstract knowledge would be stored in that machine, and a being from another star would, under certain ideal circumstances, be able to coax out elements of our knowledge from this dead entity.

Machines

The idea that technology could replicate mathematical thinking, that machines could perform and accelerate these processes, can be traced back at least as far as Pascal. There is not only his model of a small, ingenious machine; there is also a discussion of its metaphysics in the *Pensées*; a mention of it in a letter to chancellor Pierre Séguier; a cleverly composed advertising flyer; a deferential announcement to Queen Christine [of Sweden]; a royal license from the King [Louis XIV]; and finally a splendid description published in Diderot's *Encyclopédie*.[8] Leibniz designed and built his calculating machine independently from Pascal.[9] Leibniz's machine is superior to its predecessor. Pascal's license was apparently granted on 22 May 1649. Leibniz's designs date to the 1670s. He introduced the fundamental principles for a stepped reckoner (*Staffelwalze*). These principles are used in all machines that replicate mathematical thinking by purely mechanical means and perform operations by repeated addition. Such machines add sequences of numbers using a mechanical counter; each number is entered into this register by turning a crank a corresponding number of times, and the operation of multiplication is reduced to addition. All later multiplication machines proceed directly with multiplications and are generally not used to perform addition or subtraction. 'Nuremberg Scissors' and 'direct multipliers' are among the mechanical elements used by these multiplication machines. The Leibnizian idea of stepped reckoners became crucial to a number of machines built since then. It seems that two specific machines were built following his instructions. One has been lost, the other has been restored and displayed in the State Library of Lower Saxony in Hanover. Pascal's calculator was about 36 cm long, 13 cm wide, and 8 cm tall, as specified in a report to the French Academy of Sciences in 1849, in which two later engineers, Maurel and Jayet, make reference to the original inventor. Around the same time that Pascal's machine gained new traction in France, the American Babbage built a sensational calculating machine.[10] Edgar Allen Poe made it the topic of his technological and philosophical reflection in the short essay 'Maelzel's Chess-Player'.[11] This was the second time, after Diderot's magnificent description, that a calculating machine found its way into literature. Since then, various machines have come and gone. But they remain an example of modern technology that is at once real and utopian. No opportunity has been missed to build upon these

[8]Denis Diderot described Pascal's calculator [arithmetic machine] in the *Encyclopédie* (1751f).
[9]Gottfried Wilhelm Leibniz, see note 2.
[10]Charles Babbage, see note 3.
[11]Edgar A. Poe's story 'Maelzel's Chess Player' very much resonates today, considering that IBM's chess-playing computer recently defeated the world chess champion.

constructions to create the great mathematical machine, the artificial brain. The calculating machine swiftly progressed through all the stages of the technological world: the mechanical, the thermodynamic, the electrotechnical, and the high-frequency stage. Today, electrical engineering and high-frequency technology are the foundations of all major mathematical machines.

This development actually began in 1930, when the Americans Bush and Caldwell invented their famous machine[12] that could solve the differential equations that we use today to express laws of nature; they called it a 'differential analyser': its arithmetic operations were completely automated. The electrical developments that had been standard since World War I passed their first big test here. In 1942, the machine finally entered the new era of high-frequency and electrical engineering when Aiken, at Harvard University, invented his universal machine, the 'Electronic Numerical Integrator and Computer', in short: 'Eniac'.[13] Its novel system combined vacuum tube technology and punch-card (Hollerith card) techniques. This large mathematical machine represents a kind of universal calculation, insofar as it is capable of carrying out arithmetic operations such as addition, subtraction, multiplication, division, calculating integrals and differential equations, and finding square roots. What's more, it can be used to calculate tables, e.g., logarithmic tables, and can hence be programmed to produce numerical, mathematical sequences. But this may not be the most impressive part. The machine's purpose was to relieve us humans of the burden of performing these mathematical tasks, and to calculate even large numbers quickly and independently, in principle with no limitations on the number of operations. Galilei struggled to measure short falling times. He had to slow the acceleration of the falling object by using an inclined plane to observe the object's fall. Cybernetic machines can master the smallest interval. Addition takes five millionths of a second; in 5 minutes, the machine can perform 10 million additions or subtractions of 10-digit numbers. What becomes clear here is the *machine's particular relationship to time*: it works in the microstructures, the microprocesses, of time, which are inaccessible to human action and thinking. Likewise, our imagination is too limited to ever conceive of operations broken down into such infinitesimal time-frames.

How is this possible? Only a technician can grasp this. Everyone else must be content with a general reference to these principles. Their details and specific procedures are beyond the scope of this account. The machine can carry out universal operations by, for example, using pre-made punch cards. Recurring mathematical operations, such as solving particular equations, can be encoded and stored on the cards. These cards are fed into a reader, allowing the calculator to execute the mathematical operations stored on the cards. As far as the speed and volume of the calculations are concerned, the following principle applies: First and foremost, the entire technical system is based on a binary numeral system. The numbers are expressed using 2 symbols, not 10. For example, if the symbols are 0 and 1, the number 466 would be translated to the binary

[12]Vannevar Bush was a Professor at MIT (Massachusetts Institute of Technology) in Cambridge. He built the first analog computer in 1927, and developed the first electronic differential analyser in 1942. Together with S.H. Caldwell, he continued this work around the year 1930.

[13]Howard Aiken built the 'Eniac' in 1942; he developed 'Mark I' in 1939–1944 and 'Mark II' in 1947. Editor's note: Bense here misattributes the ENIAC, built by John Mauchly and J. Presper Eckert at the University of Pennsylvania, to Aiken, who developed the Mark systems at Harvard.

number 111010010. From a technical point of view, the advantage of using a base two system is that the machine only needs two numbers to represent all values. We have already seen how that can be done. Of course, this also simplifies the processes. There are essentially only two multiplications: 0 and 1. In modern machines, the two signs, 0 and 1, are realised by the leaps and gaps of a zig-zagging line that appears as the trace of the rhythms of electric pulses in certain vacuum tubes. Each number is, in other words, the result of leaps and gaps in the voltage curve, to be more precise, of the rhythm of electric pulses in each vacuum tube. Every tube which can produce such electric pulses can therefore write a set of numbers. And multiple tubes can thus write multiple sets of numbers simultaneously. It is easy to see how such an electronic calculating machine can solve operations containing large numbers, if the machine uses a combination of as many pulse generators, that is, tubes, as possible. The machine as a whole represents, as it were, a technological brain, every single tube acting essentially as a technological neuron. The tubes play their game in response to numbers, i.e., to sequences of pulses. Each reaction is stipulated by the punch card input, which contains the appropriate mathematical operation. Let me offer the following example to give an impression of the extent of this technological combination. It is common to work with high-frequency currents that generate 100,000 pulses per second. Typically, the machine contains 15,000–18,000 tubes, and tens of thousands of relays and electrical contacts. It requires over 300 kilometres of wire, and more than 100 electric motors. The entire machine can weigh up to 100 tons. In general, these machines can handle 10-digit numbers. It is furthermore possible to link two machines, which can then carry out operations with 20-digit numbers. A single tube can store over 1200 numbers, i.e., give them technological form, as signals generated at a certain time – just like images triggered by our memory in specific moments (Kafka's key!). The entire apparatus resembles the enormous, fine-grained nervous system of an intricate telephone switchboard.

The being and time of machines

There is an ontological theory of machines. Such a theory ought to develop the modes and attributes, the categories and layers of being of technical entities [*Gebilde*]. It would also show that the technical world's relation to being does not correspond to the theodicy's classical relation to being, in which things are construed as independent objects – as objects that are God's creatures and yet distant from him – which we can find in the history of philosophy from Aristotle to Christian Wolff. The machines' relation to being cannot be objectified; it cannot be reduced to a thing. The creator, the human being, is a part of this relation, but the creator remains the pivot to which everything must relate; the technical world is functional being through and through. Every machine is compossible and concessive, possible-with and necessary-with, as the ontologist says, in relation to the others. By the way, this notion in particular is the basis for the all-consuming character, in the broadest sense, of the world of machines: perfection relies on the inclusion of everything. In another context, Martin Heidegger has called such an all-consuming relation to being 'fundamental ontology' ('We are ourselves the entities to be analysed. The Being of any such entity is in each case mine', he writes in *Being and Time* [trans. Macquarrie and Robinson]). We will adopt this expression to determine the machine's relation to being.

Now, Norbert Wiener introduces a significant addition to a possible ontological theory of technological entities and aggregates. In a metaphysical chapter of his book entitled 'Newtonian and Bergsonian Time', he writes about Newton's uniformly flowing absolute time, which establishes the condition for the classical description of nature and infers that this notion of time implies a *reversible, invertible time*. Bergson's theory of 'duration' differs significantly from this Newtonian concept of time. Duration cannot be reversed. Duration is *irreversible time*. The arrow of duration [*Dauer*] always points forwards. This is, so to speak, the sense – the phenomenological sense, as Husserl would say – of an expression like 'the time of apprenticeship lasts [*dauert*] 3 years'. We can also find this little fact hidden in Proust's programmatic sentences from the overture to the novel mentioned above, which, after all, was profoundly influenced by Bergson's philosophy and his theory of duration and memory: 'And so it is with our own past. It is a labour in vain to attempt to recapture it: all the efforts of our intellect must prove futile' [trans. Moncrieff]. By the way, Walter Benjamin, who, as I have mentioned, traced the Bergsonian notions of memory in Proust's novels, alludes in his analysis to the psychoanalytic concept of memory as a destructive force. And rightly so, for memory's destructive character is tied to the irreversibility of duration, which memory aims to unravel.

If we now consider that, as Bergson has shown, duration is what distinguishes life from matter, and that the large mathematical machines, as Wiener shows, behave like human brains since they have the capacity to store intelligence signals and theorems until a future call provokes their response (if I may say so), then we should conclude that they adhere not to Newtonian *time*, but to Bergsonian *duration*. The process-oriented nature of these subtle entities requires that their being in time is not reversible, but rather irreversible. Viewed in this light, we can now look again at the temporality of Being, at the heart of Heidegger's fundamental ontology, and quickly see how the temporal dimensions of the phenomenon of *care* are equally irreversible. The fundamental ontological mode of technical entities would at least formally be compatible with Norbert Wiener's metatechnical system.

Maelzel's Chess-Player

I have already mentioned Edgar Allan Poe's essay 'Maelzel's Chess-Player', which is surely the earliest literary documentation of what we are here calling 'metatechnology'. Poe is concerned with the utopian description of an automaton that plays chess, the mechanical constitution of which allows it to react appropriately to the moves of an opponent. The profundity and perfection of Poe's technological speculation consists in the fact that Maelzel's chess-player does not always win. For that reason, Poe concludes that it is not a pure machine, and therefore immediately proceeds to the idea of a machine which wins every game against its opponent, and which would thus be constructed according to an ideal principle of game-winning in general. This is evidently the point that interests the cybernetician. The famous mathematician von Neumann occupied himself with this thought. Wiener dedicates cybernetics, metatechnical considerations to it. Such a machine would have to store all possible moves, like the numbers in the individual tubes of the ENIAC. And this storage would represent nothing other than what we might call the extreme concentration of the chess player, who is also in possession of all moves and countermoves, i.e., who holds these moves in storage. The concentrated

memory conveys to us a notion of what the philosopher occasionally calls the 'unity of consciousness', or even the 'unity of the I'. Every tube would represent such a 'unity of consciousness' in technical form, which is the precondition for its ability to respond reasonably to signals, to the moves of the opponent. What the metaphysician characterises in his language as the transcendental-logical unity of the I, as Gotthard Günther has expressed it,[14] corresponds for the metatechnologist to the pulse behaviours of the tubes, which function according to plan, storing and reacting. Thus there would be a future for Maelzel's chess-player, which can lose games, as well as for Poe's chess-player, which cannot, in the realm of mathematical and technical expertise. Maelzel's automaton allows cheating, both theoretically and practically; this is why it can never be a pure machine, as Poe says, and still reveals the indeterminate contour of life. But Poe's automaton, which wins every game, is a pure machine. It represents a completely closed, fully knowable [*übersehbare*] unit of intelligence storage, a gapless clockwork [*Automatik*] which, as Wiener would point out, is not evil, but ridiculous. This line of thought shows that precision within the completely closed unity of a reactive consciousness can only be achieved by the sacrifice of error, of freedom, and therefore of life. The machine may outstrip the detailed spectrum of actual human knowledge, but it nevertheless only knows what we can know without knowing that we know it, as H. E. Hollmann has recently put it.[15] Maelzel's chess-player – which is not a complete machine, since it does not possess the principle that underlies all winning moves, or consciously does not employ this principle (Poe leaves this a bit open) – reminds me of the 'evil demon' of Descartes' *Meditations*. Recall Descartes' line of thought: because God cannot want me to be mistaken about that which is fully self-evident – for this would contradict his omnipotence, his goodness, his love of truth – the impossibility of any error in the completely personal self-certitude of the I provides the basis for the proof of truths about everything in existence. The indivisible, transcendental, I is always presumed as the starting point, and so long as this I – since God is incapable of deception – is at least in principle in a position to think correctly, it demonstrates space and time and world. The complete, knowable, closed unity of the I is a precondition of any precise calculation; the truth inside which there can be no error, or the game in which there can be no defeat, follows from the coordination of the two. The technological unity of a machine and the transcendental unity of an I can be seen equally well from this Cartesian perspective. It would appear that, in this way, the greatest clarity about the metatechnological, platonic essence of the machine can be achieved.

Metatechnology and anthropology

We have described the path from Galileo's physics to cybernetics as a process of the steady dismantling of nature on the one hand, but also of natural reason on the other.

[14]Gotthard Günther published his book *Das Bewußtsein der Maschinen* [*The Consciousness of Machines*] in 1958 with the Agis-Verlag in Baden-Baden, and it appeared in an expanded second edition in 1963. Similar considerations can be found in Günther's *Beiträge zur Grundlegung einer operationsfähigen Dialektik* [*Contributions to the Foundation of an operational Dialectics*], volume 1, Hamburg 1976.

[15]Hans Erich Hollmann founded an electromedical laboratory in Berlin that was tasked with armament duties. Max Bense worked as a physicist and mathematician from 1942 to 1945 in this laboratory. Hollmann went to America after the war, where he worked for NASA in California.

Mathematical symbol systems increasingly took the place of the original Word, just as technically constructed processes took the place of natural processes. The result is technical intelligence [*Intelligenz*], which has slowly entrenched us in a technical world. Intelligence and world condition each other, and that is just as much a cybernetic proposition as an anthropological one. Thus one can legitimately ask about the role of metatechnology in the context of philosophical anthropology, which has a metaphysical character in Scheler, an ontological one in Heidegger, and an empirical one in Arnold Gehlen[16] (*Der Mensch, seine Natur und seine Stellung in der Welt*, 1940; *Man: His Nature and Place in the World*, 1988). The mismatch between nature and the human that all of these thinkers admit extends from Nietzsche to Herder to Pascal, among others, and far further back. We can assume it as a premise. The conclusion that the human cannot be completely and clearly explained from nature, that the human, as Nietzsche expresses it, is the 'not yet determined animal', is an important starting point of modern anthropology. The human is forced, since technical phenomena appear to be penetrating into the deeper layers of fundamental human relations [*Seinsverhältnisse*] – as cybernetic machines indeed have made clear – to view technology as a possible solution for this anthropologically definable mismatch. Using technology, the human creates an *environment* for himself, which corresponds to his double role as nature-bound and intellectual being. The technical world is an environment, a being-like sphere from which we can completely explain, at least ideally, what we call technical existence and technical intelligence. In the nature-bound world, *the human*, as it was formulated, *was the higher but also weaker being*; in the technical world, he is the stronger creature through and through. We do not say that the technical world is the only attempt that the human has made so far to *set himself up in nature*, but it appears it is the most systematic and perhaps most successful of the attempts that have so far proven possible to act in such a way as to compensate for this evident mismatch. Manual, physical processes have technical potential in the same sense as intelligible and psychological processes. Abstract mathematical calculating systems show this just as much as cybernetic machines. It follows that under this limited lens, the sphere of the technical is more comprehensive than the sphere of what we have called nature or spirit. Technical being comprehends both. That is the reason that a being like the human, which is more or less both natural and spiritual, possesses, as the inhabitant of the technical world, a chance to justify his existence in terms of being. *The human as technical existence*: this seems to me one of the great tasks of a philosophical anthropology of tomorrow.

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Max Bense (1910-1990) was a German writer and philosopher whose work frequently addressed the intersection of technology, ontology, and aesthetics. He taught at the Technische Hochschule in Stuttgart (renamed the University of Stuttgart in 1967) from 1949 until his retirement in 1978.

[16]Arnold Gehlen's book *Der Mensch, seine Natur und seine Stellung in der Welt* was published in 1940. All following editions have been expanded.

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